

Lab 1: Basic Unix Commands

Question 1:

Suppose, you are in your home directory (i.e., /home/EASTERN/*your_user_id*).

- a) What will be printed by **pwd** after you run the following command?
cd ../../..

It will print `/` representing the root directory.

```
jblazina@cscd-linux01: /  
jblazina@cscd-linux01:~$ pwd  
/home/EASTERN/jblazina  
jblazina@cscd-linux01:~$ cd ../../..  
jblazina@cscd-linux01:/$ pwd  
/  
jblazina@cscd-linux01:/$
```

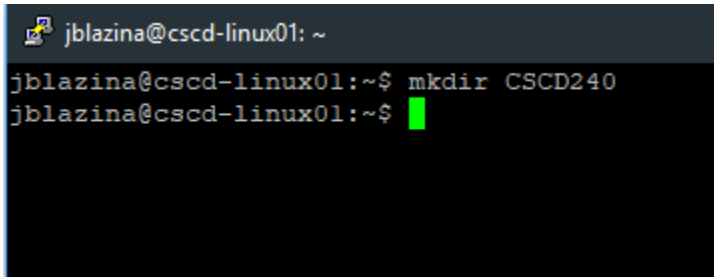
- b) What command will bring you back to your home directory?

cd ~

```
jblazina@cscd-linux01: ~  
jblazina@cscd-linux01:~$ pwd  
/home/EASTERN/jblazina  
jblazina@cscd-linux01:~$ cd ../../..  
jblazina@cscd-linux01:/$ pwd  
/  
jblazina@cscd-linux01:/$ cd ~  
jblazina@cscd-linux01:~$ pwd  
/home/EASTERN/jblazina  
jblazina@cscd-linux01:~$
```

- c) Under your home directory, create a directory named **CSCD240**

mkdir CSCD240

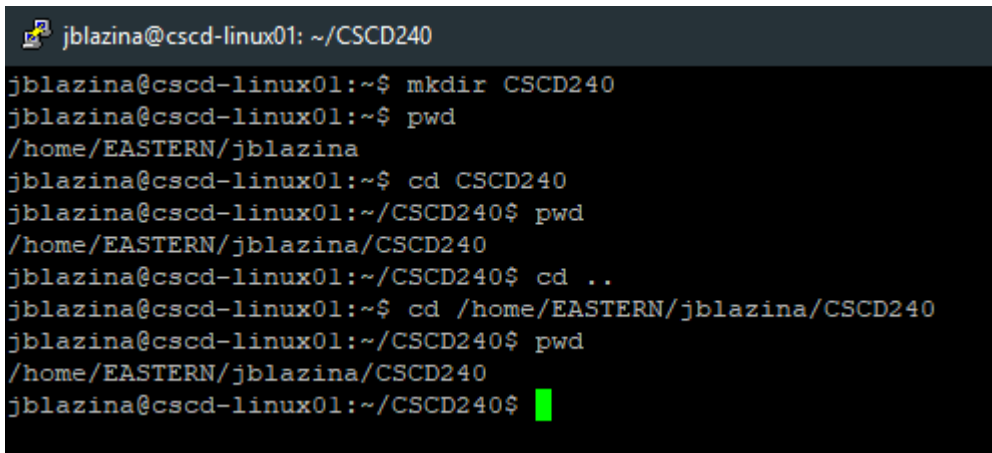
A terminal window with a dark background. The prompt is 'jblazina@cscd-linux01: ~'. The user enters 'mkdir CSCD240' and the prompt returns. A green cursor is visible at the end of the second prompt line.

```
jblazina@cscd-linux01: ~  
jblazina@cscd-linux01:~$ mkdir CSCD240  
jblazina@cscd-linux01:~$
```

- d) You want to access the directory named **CSCD240** from your home directory. Write down both the **absolute path** and **relative path** for that

Absolute: cd /home/EASTERN/jblazina/CSCD240

Relative: cd CSCD240

A terminal window with a dark background. The prompt is 'jblazina@cscd-linux01: ~/CSCD240'. The user enters a series of commands: 'mkdir CSCD240', 'pwd' (returns '/home/EASTERN/jblazina'), 'cd CSCD240', 'pwd' (returns '/home/EASTERN/jblazina/CSCD240'), 'cd ..', 'cd /home/EASTERN/jblazina/CSCD240', 'pwd' (returns '/home/EASTERN/jblazina/CSCD240'), and 'pwd' (returns '/home/EASTERN/jblazina/CSCD240'). A green cursor is visible at the end of the last prompt line.

```
jblazina@cscd-linux01: ~/CSCD240  
jblazina@cscd-linux01:~$ mkdir CSCD240  
jblazina@cscd-linux01:~$ pwd  
/home/EASTERN/jblazina  
jblazina@cscd-linux01:~$ cd CSCD240  
jblazina@cscd-linux01:~/CSCD240$ pwd  
/home/EASTERN/jblazina/CSCD240  
jblazina@cscd-linux01:~/CSCD240$ cd ..  
jblazina@cscd-linux01:~$ cd /home/EASTERN/jblazina/CSCD240  
jblazina@cscd-linux01:~/CSCD240$ pwd  
/home/EASTERN/jblazina/CSCD240  
jblazina@cscd-linux01:~/CSCD240$
```

- e) After running the command in **d)**, what command will you use to figure out your current working directory?

pwd

(SEE ABOVE SCREENSHOT FROM QUESTION 1.d)

- f) Create a file named **Assignment1** under **CSCD240**.

touch Assignment1

```
jblazina@cscd-linux01:~/CSCD240$ touch Assignment1
jblazina@cscd-linux01:~/CSCD240$ ls
Assignment1
jblazina@cscd-linux01:~/CSCD240$
```

- g) Create a hard link for the file named **Assignment1** in your current directory. You can choose any name.

ln Assignment1 hardlink

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ ln Assignment1 hardlink
jblazina@cscd-linux01:~/CSCD240$ ls
Assignment1 hardlink
jblazina@cscd-linux01:~/CSCD240$
```

- h) Suppose you are now in **CSCD240** folder. What output will be produced by the following command? Explain.

ls ~ .

It will list all the contents of the home directory

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ ls ~
CSCD240 netstorage OLD
jblazina@cscd-linux01:~/CSCD240$
```

- i) You need to copy **Assignment1** file from your current directory (**CSCD240**) to your home directory. What command will you use?

cp Assignment1 ~

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ cp Assignment1 ~
jblazina@cscd-linux01:~/CSCD240$ ls ~
Assignment1  CSCD240  netstorage  OLD
jblazina@cscd-linux01:~/CSCD240$
```

- j) Create one folder named **Assignment** under **CSCD240**. Move the file named **Assignment1** from current folder (**CSCD240**) to **Assignment** folder. What commands will you use?

mkdir Assignment

mv Assignment1 Assignment

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ mkdir Assignment
jblazina@cscd-linux01:~/CSCD240$ mv Assignment1 Assignment
jblazina@cscd-linux01:~/CSCD240$ ls
Assignment  hardlink
jblazina@cscd-linux01:~/CSCD240$
```

- k) Copy the **Assignment** folder from current folder (CSCD240) to your home directory.
What command will you use?

cp -r Assignment ~

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ cp -r Assignment ~
jblazina@cscd-linux01:~/CSCD240$ ls ~
Assignment Assignment1 CSCD240 netstorage OLD
jblazina@cscd-linux01:~/CSCD240$
```

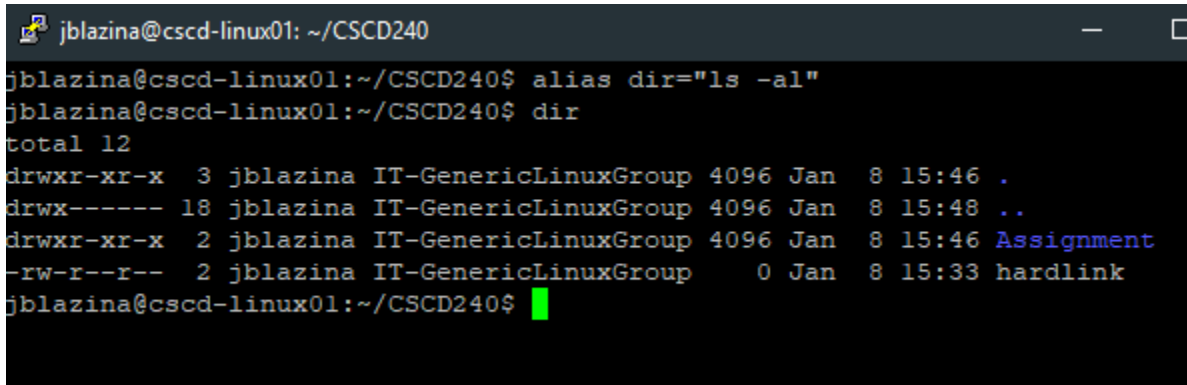
- l) What is the difference between 'ls -l' and 'ls -al' commands?

ls -l will show NOT show hidden content, ls -al WILL show hidden content.

```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ ls -l
total 4
drwxr-xr-x 2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 Assignment
-rw-r--r-- 2 jblazina IT-GenericLinuxGroup    0 Jan  8 15:33 hardlink
jblazina@cscd-linux01:~/CSCD240$ ls -al
total 12
drwxr-xr-x  3 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 .
drwx----- 18 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:48 ..
drwxr-xr-x  2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 Assignment
-rw-r--r--  2 jblazina IT-GenericLinuxGroup    0 Jan  8 15:33 hardlink
jblazina@cscd-linux01:~/CSCD240$
```

- m) Make a new command **dir** that is equivalent to unix command **ls -al**. Capture the screenshot of the command that can achieve that and the results.

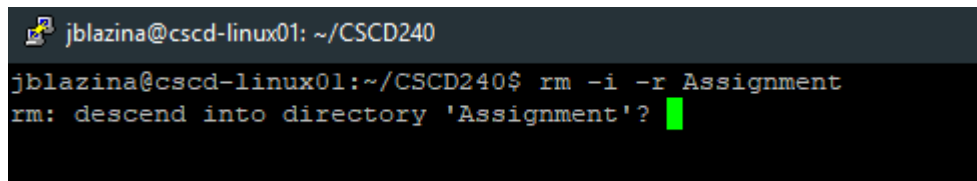
Alias dir="ls -al"



```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ alias dir="ls -al"
jblazina@cscd-linux01:~/CSCD240$ dir
total 12
drwxr-xr-x  3 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 .
drwx----- 18 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:48 ..
drwxr-xr-x  2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 Assignment
-rw-r--r--  2 jblazina IT-GenericLinuxGroup   0 Jan  8 15:33 hardlink
jblazina@cscd-linux01:~/CSCD240$
```

- n) We'd like to get a warning or prompt information before we delete the subdirectory **Assignment** under **CSCD240**. What command will you use?

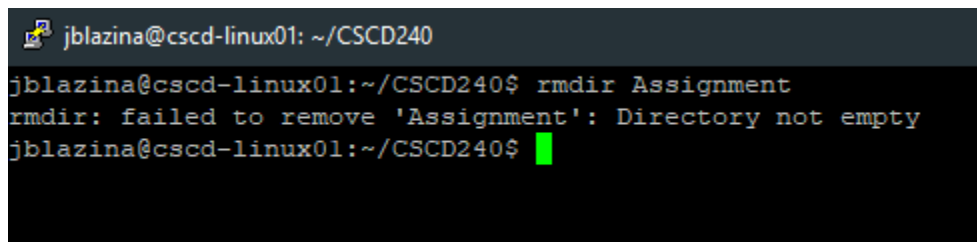
rm -i -r Assignment



```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ rm -i -r Assignment
rm: descend into directory 'Assignment'?
```

- o) Use **rmdir** to delete subdirectory **Assignment** under **CSCD240**. Does it delete the directory? Why or why not?

It does not delete the subdirectory because it is not empty.



```
jblazina@cscd-linux01: ~/CSCD240
jblazina@cscd-linux01:~/CSCD240$ rmdir Assignment
rmdir: failed to remove 'Assignment': Directory not empty
jblazina@cscd-linux01:~/CSCD240$
```

Question 2. Suppose you are in your home directory.

- a) Create a text file named **calendar2019.txt** using command **cal 2019 > calendar2019.txt**. Issue the **more** command or the **less** command on **calendar2019.txt** and capture the screenshot of the output. How to move to the beginning of **calendar2019.txt** in **less**? How to move to the end of **calendar2019.txt** in **less**? How to scroll down or up?

g: Moves to Beginning

G: Moves to End

j: Scroll Down

k: Scroll Up

```
jblazina@cscd-linux01: ~
jblazina@cscd-linux01:~$ cal 2019 > calendar2019.txt
jblazina@cscd-linux01:~$ less calendar2019.txt

      2019
  January  February  March
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4  5          1  2          3  4  5  6  7  8  9
  6  7  8  9 10 11 12    3  4  5  6  7  8  9    10 11 12 13 14 15 16
13 14 15 16 17 18 19    10 11 12 13 14 15 16    17 18 19 20 21 22 23
20 21 22 23 24 25 26    17 18 19 20 21 22 23    24 25 26 27 28 29 30
27 28 29 30 31          24 25 26 27 28          31

  April      May      June
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4  5  6          1  2  3  4          2  3  4  5  6  7  8
  7  8  9 10 11 12 13    5  6  7  8  9 10 11    9 10 11 12 13 14 15
14 15 16 17 18 19 20    12 13 14 15 16 17 18    16 17 18 19 20 21 22
21 22 23 24 25 26 27    19 20 21 22 23 24 25    23 24 25 26 27 28 29
28 29 30          26 27 28 29 30 31          30

  July      August      September
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa
      1  2  3  4  5  6          1  2  3          1  2  3  4  5  6  7
  7  8  9 10 11 12 13    4  5  6  7  8  9 10    8  9 10 11 12 13 14
14 15 16 17 18 19 20    11 12 13 14 15 16 17    15 16 17 18 19 20 21
21 22 23 24 25 26 27    18 19 20 21 22 23 24    22 23 24 25 26 27 28
28 29 30 31          25 26 27 28 29 30 31    29 30

  October      November      December
```

- b) Remove **read permission** from **calendar2019.txt** for the owner and **all permissions** for the group. Write down the commands using both symbolic and numeric (octal) values. Capture the screenshot for the commands and prove that the permission was changed.

Symbolic: **chmod u-r calendar2019.txt**
 chmod g-rwx calendar2019.txt

Octal: **chmod 307 calendar2019.txt**

```
jblazina@cscd-linux01: ~  
jblazina@cscd-linux01:~$ ls -l  
total 16  
drwxr-xr-x  2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:48 Assignment  
-rw-r--r--  1 jblazina IT-GenericLinuxGroup   0 Jan  8 15:45 Assignment1  
-rw-r--r--  1 jblazina IT-GenericLinuxGroup 2184 Jan  8 16:12 calendar2019.txt  
drwxr-xr-x  3 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 CSCD240  
lrwxrwxrwx  1 jblazina IT-GenericLinuxGroup  16 Apr  2  2018 netstorage -> /mnt  
lazina  
drwxr-xr-x 15 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:43 OLD  
jblazina@cscd-linux01:~$ chmod u-r calendar2019.txt  
jblazina@cscd-linux01:~$ chmod g-rwx calendar2019.txt  
jblazina@cscd-linux01:~$ ls -l  
total 16  
drwxr-xr-x  2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:48 Assignment  
-rw-r--r--  1 jblazina IT-GenericLinuxGroup   0 Jan  8 15:45 Assignment1  
--w----r--  1 jblazina IT-GenericLinuxGroup 2184 Jan  8 16:12 calendar2019.txt
```

```
jblazina@cscd-linux01: ~  
jblazina@cscd-linux01:~$ chmod 307 calendar2019.txt  
jblazina@cscd-linux01:~$ ls -l  
total 16  
drwxr-xr-x  2 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:48 Assignment  
-rw-r--r--  1 jblazina IT-GenericLinuxGroup   0 Jan  8 15:45 Assignment1  
--wx---rwx  1 jblazina IT-GenericLinuxGroup 2184 Jan  8 16:12 calendar2019.txt  
drwxr-xr-x  3 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:46 CSCD240  
lrwxrwxrwx  1 jblazina IT-GenericLinuxGroup  16 Apr  2  2018 netstorage -> /mnt  
lazina  
drwxr-xr-x 15 jblazina IT-GenericLinuxGroup 4096 Jan  8 15:43 OLD  
jblazina@cscd-linux01:~$
```


Question 3. Explain the following outputs from a unix command:

```
-rw-r--r-- 1 sysadmin IT-GenericLinuxGroup 3637 Sep 21 2015 file.txt
```

Likely Command: `ls -l`

Explanation: Shows information of a file, this consists of a file.txt owned by sysadmin that was last modified Sep 21 2018 and is in the IT-GenericLinuxGroup group, with 1 link.

Permissions: **User – read/write**

Group – read

Others - read

```
drwx----- 16 sysadmin IT-GenericLinuxGroup 4096 Mar 29 2016 CSCD240
```

Command: `ls -l ~` (most likely from home directory based on previous contents of this lab)

Explanation: Shows visible contents of a directory, this directory consists of a subdirectory CSCD240 owned by sysadmin that was last modified Mar 29 2016 and is in the IT-GenericLinuxGroup group, with 16 links.

Permissions: **User – read/write/execute**

Group – none

Others - none

```
lrwxrwxrwx 1 sysadmin IT-GenericLinuxGroup 15 Nov 21 2015 netstorage -> /mnt/ns-sysadmin
```

Explanation: Shows a symbolic link owned by sysadmin that was last modified Nov 21 2015 and is in the IT-GenericLinuxGroup group, with 1 links.

Permissions: **User – read/write/execute**

Group – read/write/execute

Others – read/write/execute

```
-rw-r--r-- 2 sysadmin IT-GenericLinuxGroup 80 Jan 24 2017 hello.c
```

Explanation: Shows file information regarding hello.c owned by sysadmin that was last modified Jan 24 2017 and is in the IT-GenericLinuxGroup group, with 2 links.

Permissions: **User – read/write**

Group – read

Global – read